

Nasim Beigzadeh

Razi University, Kermanshah, Iran
M.Sc. in Artificial Intelligence
Graduate Research Assistant

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Summary

Researcher with **4+ years** of experience at the Signal and Image Processing Laboratory, specializing in **medical imaging, biomedical signal processing, and deep learning for healthcare applications**. Experienced in designing lightweight and computationally efficient deep learning architectures through **knowledge distillation and model compression**, applied to cardiovascular disease diagnosis from ECG signals, with performance suitable for low-power and resource-constrained healthcare devices. Author of **four journal papers extracted from my M.Sc. thesis**. Proficient in Python, PyTorch, TensorFlow, and OpenCV. Seeking Ph.D. research opportunities in **artificial intelligence and deep learning**, with broader interests spanning medical applications, computer vision, and intelligent systems.

Education

M.Sc. in Artificial Intelligence, Computer Engineering 2021–2024
Razi University, Kermanshah, Iran

- Thesis: *Diagnosis of Cardiac Diseases by Applying a Combination of Knowledge Distillation and Transformer Model on ECG Signals*.
- Supervisor: [Prof. Abdolhossein Fathi](#).
- Thesis Grade: A⁺
- Total CGPA: **3.92/4.00** (WES iGPA)

B.Sc. in Hardware Engineering, Computer Engineering 2014–2020
University of Science and Technology of Mazandaran, Mazandaran, Iran

- Thesis: *Design and Implementation of a Smart Irrigation System Using Arduino*.
- Thesis Grade: A⁺

Research Interests

- Imaging & Signal Processing
- Image Segmentation
- Lightweight Neural Network Design
- Machine Learning & Deep Learning
- Neuromorphic Spiking Neural Networks
- IoT & AI-driven Intelligent Systems

Technical Skills

Programming Languages: Python, Arduino (basic)

Deep Learning Frameworks: PyTorch, TensorFlow, Keras

Scientific Libraries: NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn

Computer Vision: OpenCV, image preprocessing, and augmentation techniques

Tools: Git, GitHub, CUDA-based GPU Computing, LaTeX

Publications

Journal articles
(Published)

- **Beigzadeh, N., & Fathi, A. (2026).** [HMT-KD: Lightweight Deep Learning Model for ECG Analysis Using Hierarchical Multi-Teacher Knowledge Distillation Framework](#). **Applied Soft Computing** (Q1, IF ≈ 7.8).
- **Beigzadeh, N., & Fathi, A. (2026).** [Enhancing Cardiovascular Disease Diagnosis through Knowledge Distillation and Hybrid Transformer–CNN–SENet Architecture](#). **Journal of Medical Engineering & Technology**.
- **Beigzadeh, N., & Fathi, A. (2025).** [Employing a Deep Learning-based Model on Printed Images of ECG Signals for Cardiovascular Disease Detection](#). **Journal of Machine Vision and Image Processing**.

Under Review

- **Beigzadeh, N., Shahsavari, M., & Fathi, A. (2026).** EIA-CKD: An End-to-End Internal Augmentation and Curriculum-Based Knowledge Distillation Framework for Lightweight ECG Classification. **Biomedical Signal Processing and Control** (Q1, IF ≈ 5.7).
- Salehnia, T., **Beigzadeh, N.**, & Fathi, A. (2026). Natural Image Segmentation Using Meta-heuristic Algorithms.

Research Experience	Research Collaborator 2025–Present <i>Donders Institute for Brain, Cognition and Behaviour, Radboud University, The Netherlands</i> Supervisor: Dr. Mahyar Shahsavari Co-supervisor: Prof. Abdolhossein Fathi Developed EIA-CKD, a lightweight curriculum-based knowledge distillation framework for cardio-vascular disease classification from ECG images, evaluated on the PTB-XL and Chapman datasets, targeting efficient diagnosis for resource-constrained and wearable healthcare systems.
Teaching Experience	Graduate Research Assistant Sep 2023–Jul 2025 <i>Signal and Image Processing Laboratory (SIPL), Razi University, Kermanshah, Iran</i> Supervisor: Prof. Abdolhossein Fathi - Conducted research in medical imaging, biomedical signal processing, and deep learning. - Developed lightweight deep learning architectures for cardiovascular disease diagnosis using ECG signals. - Worked extensively on knowledge distillation and model compression for computationally efficient neural networks. - Designed hybrid CNN–SENet–Transformer architectures for ECG classification. - Investigated efficient deep learning models suitable for low-power and resource-constrained healthcare devices. Teaching Assistant – Neural Networks Jan 2025–Jun 2025 <i>Department of Computer Engineering, Razi University</i> Instructor: Dr. Abdollah Chalechale - Assisted in teaching and evaluating coursework related to neural network architectures. - Held weekly tutorial sessions and guided students on implementation projects using PyTorch, NumPy, and Pandas.
Educational Activities	Instructor – Neural Network Functions in Python & Deep Learning for ECG Signal and Motion Analysis 2025 <i>Farazan Knowledge-Based Startup Center, Razi University Science and Technology Park</i> Under the supervision of Dr. Abdollah Chalechale.
Peer Review Experience	- Completed 14 reviews for Knowledge-Based Systems (Q1). - Completed 1 review for Applied Soft Computing (Q1). - Completed 1 review for Biomedical Signal Processing and Control (Q1). - Peer-review activity conducted through ORCID: ORCID Profile.
Selected Technical Projects	- Medical image segmentation using U-Net - Knowledge distillation for model compression - ECG image generation using GANs - Hybrid CNN–Transformer model optimization for ECG - Fine-tuning ResNet, DenseNet, and related architectures for diagnostic imaging - RNN/LSTM modeling for temporal ECG classification - Neural Architecture Search (NAS) for optimized network design
Honors & Achievement	Ranked 3rd of graduating class among MSc students, Artificial Intelligence, Department of Electrical and Computer Engineering, Razi University 2024
Languages	English: TOEFL iBT (Nov. 2026) Persian: Native Kurdish: Native
References	Prof. Abdolhossein Fathi Full Professor, Department of Computer Engineering and Information Technology, Razi University a.fathi@razi.ac.ir Dr. Abdollah Chalechale Associate Professor, Department of Computer Engineering and Information Technology, Razi University chalechale@razi.ac.ir Dr. Mahyar Shahsavari Assistant Professor, Donders Institute for Brain, Cognition and Behaviour, Radboud University mahyar.shahsavari@donders.ru.nl